

Type of the Paper (Case Study)

Title: 워터백을 적용한 균형 운동프로그램이 20대 남자대학생의 하지안정성 및 뇌파에 미치는 영향 (Effect of Water Bag applied Balance Exercise Program on Lower Body Stability and EEG in 20's Male College Students)

Abstract: This study was conducted to find out how acute balance exercise using a water bag, dumbbell, and medicine ball affects lower body stability and EEG. It was conducted on 10 male C college students in their 20s. The balance exercise program was carried out at intervals of seven days using three exercise equipment, 5kg water bag, 5kg medicine ball and two 2.5kg dumbbells, and measured Y-balance test, foot pressure and EEG before and after exercise. The results of this study. First, as a result of analyzing the Y balance test, statistically significant improvement was found in the left foot anterior, left posterior lateral, left posterior medial, right posterior lateral, and right posterior medial, followed by dumbbell, medicine ball, and water bag. Second, in the case of static foot pressure, all three were significantly improved compared to before exercise only in the surface left foot. Third, there was no significant difference in the study on EEG before and after balance exercise. In conclusion, it is thought that acute balance exercise affects lower body stability of male college students in their 20s. Although the effect of water bag on lower body stability was not fully identified in this study, statistics showed that water bag and other exercise equipment showed the effect of balanced motion. In the future, follow-up studies on various subjects and long-term exercise programs should be carried out.

Keywords: water bag, balance exercise, lower body stability, EEG, 20's Male College Student